

Diving Emergency Action Plan

Dive Park, Beaver Lake Arkansas

Emergency Phone Numbers

Divers Alert Network 1-919-684-9111
Dive Rescue International 1-800-248-3483
IADRS 1-800-IADRS-911
Mercy Hyperbaric and Wound Care 1-479-338-3550
Mercy Emergency Department 1-479-338-8000
Beaver Dam Dive Shop and Resort 479-253-9799

Nearest Recompression Facility

Mercy Hyperbaric and Wound Care
2708 S Rife Medical Ln Suite T40, Rogers, AR 72758
1-479-338-3550
Distance: ~31 miles
Travel Time: ~45 minutes

Nearest Diving Facility

Beaver Dam Dive Shop and Resort
11792 AR-187, Eureka Springs, AR 72631
1-479-253-9799
Distance: ~3 miles
Travel Time: ~6 minutes

Checklist

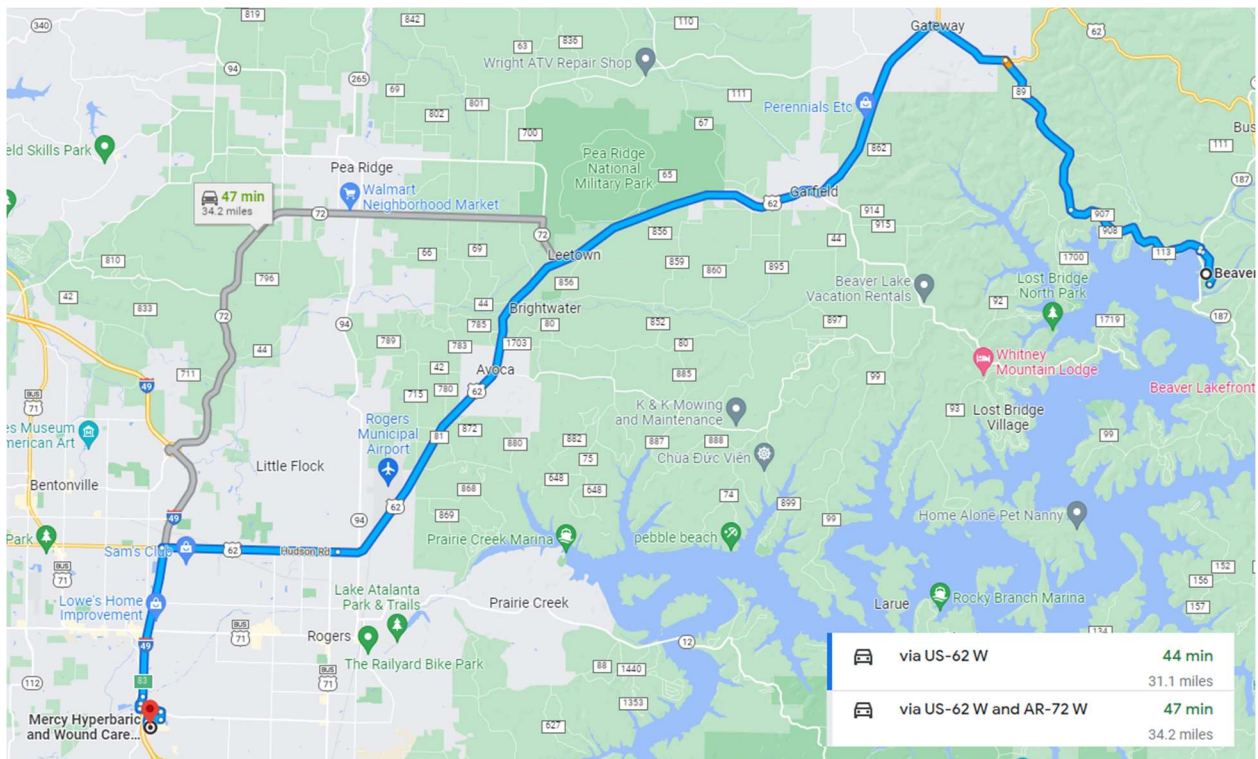
- ☐ Ambulance for Transport?
 - ☐ On Site
 - ☐ By Radio or Telephone
- ☐ Emergency Equipment in Vicinity
 - REQUIRED**
 - ☐ AED or other Cardiac Defibrillator
 - ☐ Oxygen Delivery Unit
 - ☐ First Aid Kit
 - ☐ Communication Equipment
 - PREFERRED**
 - ☐ Advanced Life Support Equipment
- ☐ Identify Trained EMS Providers in the Class
 - ☐ Paramedics _____
 - ☐ EMTs _____
 - ☐ Other _____

Action Plan

- ☐ Recall ALL divers, conduct head count
- ☐ Notify dispatch
- ☐ Request additional resources (as needed)
- ☐ Stabilize the patient/incident
- ☐ Conserve property (as appropriate)
- ☐ Turn patient/scene over to the authority having jurisdiction
- ☐ Pass on information (as needed)
- ☐ Document the incident
- ☐ Obtain statements from witnesses (as needed)
- ☐ Notify Supervisor

Diving Emergency Action Plan

Reference Material



Diving Emergency Action Plan

Accident Management (1/2)

BASIC STEPS FOR MANAGING DIVING ACCIDENTS

1. **Stop everything** going on and try to think clearly
2. **Assess the scene** of the incident for hazards. If the scene appears unsafe, do not proceed with rescue efforts. Call 9-1-1. If the scene can be made safe while ensuring the safety of all rescuers, proceed with rescuing the injured diver.
3. **Use any pertinent resources available** to you. If you need others to help, direct them with clear and concise instructions. People will follow a leader's instruction in stressful times. Ask for help if you need.
4. **Exposure protection** – Use nitrile gloves and a face mask if bodily fluids are present. These items are in the emergency dive first aid kit.
5. **Check the victim** for level of responsiveness using the acronym AVPU.
 - a. A – Injured diver is aware and oriented to him/herself, the time, and the location.
 - b. V – Injured diver gives a response when presented with a verbal stimulus.
 - c. P – Injured diver gives a response only when presented with a painful stimulus.
 - d. U – Injured diver is completely unresponsive and shows no mental function.
6. **Call for help - 9-1-1.** If the injured diver is not A on AVPU scale or if there is any question whether EMS may be needed, call 9-1-1.
 - a. If appropriate, call the Diver's Alert Network for consultation and advice.
 - b. Call the DSO or other DCB members for help and further guidance
7. **Care for the victim.** Always prioritize ABC's – airway, breathing, and circulation. If the victim is unconscious, start your care with Look-Listen-Feel for both breathing and the presence of a pulse. If a pulse is present without breathing, rescue breaths are appropriate. If neither pulse nor breathing are present, initiate CPR.
 - a. Position the injured diver on his/her back.
 - b. Give 30 compressions about 2" deep on the center of the sternum at a rate of 100-120 compressions per minute.
 - c. Open victim's airway with head-tilt, chin-lift method.
 - d. Provide two breaths to the victim.
 - e. Continue this pattern of 30 compressions to 2 breaths for 2 minutes (about 5 sets of 30:2).
 - f. Look, listen, and feel for signs of life every
 - g. Continue this process as long as possible.
8. **Oxygen.** Generally, the best thing for the majority of diving accidents is administration of 100% Oxygen. The first choice of oxygen delivery equipment for any injured diver is the demand regulator included in the oxygen kit. Second choice is the non-resuscitator bag with the oxygen hook up.

Diving Emergency Action Plan Accident Management (2/2)

9. **Treat for shock** by:
 - a. Maintaining the diver's temperature.
 - b. If possible, place the injured diver supine with legs elevated 10-12" (if head, neck, or back injury is not suspected). If this is not possible, place diver in a position of comfort.
 - c. Provide oxygen.
 - d. Do not administer fluids orally.
10. **Notes.** If there are enough people present, assign someone to take notes on the patient's state. Note times of readings and/or changes in symptoms. Write clearly. This information should be handed off to paramedics when they arrive on the scene and should stay with the patient.
11. **Accident reports.** Each diver at the scene should fill out an accident report while details are fresh in their minds. As time passes, many details often slip. Please be honest and accurate as to what you saw you. Each diver is to submit their own story and to not corroborate stories with others. Accident reports will be submitted to the Dive control board within 24 hours of the incident.
12. **Dive gear.** Do not disassemble the dive gear of the injured diver in the case of a serious accident. Please keep in-tact and label with the diver's name. The dive gear will be passed on to the DSO and the DCB.

Diving Emergency Action Plan

Dive Accident Notes

Patient's Name: _____ Date/Time: _____

Describe pain/numbness: _____

HISTORY

Number of dives in past 24 hours: _____ Depth of last dive(ft): _____ Duration: _____

Symptoms notice before, during, or after the dive: _____

If during the dive, was it while descending, at depth, or ascending: _____

Symptoms increased or decreased since first noticed: _____

Other symptoms since onset? Describe: _____

Experienced similar symptoms before: _____

Ever previously had DCS or air embolism: _____ If Yes, When: _____

HISTORY

Time	Pulse (beats/min)	Breathing (breaths/min)	AVPU (Alert, Verbal, Pain, Unresponsive)

Medications: _____ Allergies: _____

Time started on Oxygen: _____ CPR started: _____ CPR ended: _____

Calls placed (Check the appropriate boxes): ☐ 9-1-1 ☐ DAN ☐ DSO ☐ DCB ☐ Hospital

Diver's Alert Network suggestions for treatment: _____

Data recorded by _____ Contact Phone: _____

Diving Emergency Action Plan

LUNG OVER-EXPANSION INJURIES

ARTERIAL GAS EMBOLISM (AGE)

As a diver surfaces without exhaling, air trapped in the lungs expands and may rupture lung tissue releasing gas bubbles into the circulatory system where they may be distributed to the body tissues. The ascending diver is normally in a vertical position and the bubbles tend to travel upward toward the brain, eventually reaching a small artery blocking circulation. The effects of halting circulation to the brain are critical and require immediate treatment. Symptoms of embolism may be present when the victim reaches the surface or within a few minutes afterwards.

CAUSES

- Holding breath during ascent while breathing compressed air
- Lung disease causing air trapping
- Diving with cold or chest congestion
- Airway obstruction from foreign object in the mouth (gum, etc.)

SYMPTOMS

- Unconsciousness within five minutes of surfacing
- Dizziness or staggering
- Visual disturbances
- Paralysis
- Bloody froth from the mouth or nose
- Respiratory arrest

NOTE: Symptoms usually appear within 15 minutes of surfacing from a dive

TREATMENT

- EVACUATE TO RECOMPRESSION CHAMBER/HOSPITAL ASAP!
- ABC's - airway, breathing, and circulation
- Administer 100% oxygen with diver in supine position
- Administer fluids (water) to conscious patients

Diving Emergency Action Plan

LUNG OVER-EXPANSION INJURIES

PNEUMOTHORAX

As a diver ascends, the lungs typically vent expanding air without problem. If the air is blocked from exiting normally, the lungs can overinflate and damage the alveoli and bronchial passages. If expanding air ruptures, the lung, air escapes into the small, normally airless area between the lungs and chest. This is a pneumothorax. The two types of pneumothoraxes include: 1) simple pneumothorax - involves a onetime event of air escaping from the lungs into the pleural cavity AND 2) tension pneumothorax involves a repeated leaking of air into the pleural cavity with each breath. Both are very serious and require immediate medical attention.

CAUSES

- Holding breath during ascent while breathing compressed air
- Lung disease causing air trapping
- Diving with cold or chest congestion
- Airway obstruction from foreign object in the mouth (gum, etc.)

SYMPTOMS

- Difficulty or rapid breathing
- Shortness of breath
- Hypotension
- Cyanosis and shock
- Chest pains (deep breath hurts) and intense pressure in the chest
- Leaning towards affected side (and absent lung sounds on affected side)
- Loss of consciousness or death

TREATMENT

- EVACUATE TO HOSPITAL ASAP!
- ABC's - airway, breathing, and circulation
- Administer 100% oxygen with diver in supine position
- Position patient on injured side
- Treat for shock
- In the case of a tension pneumothorax, air must be vented from the chest cavity by a medical professional

Diving Emergency Action Plan

LUNG OVER-EXPANSION INJURIES

SUBCUTANEOUS EMPHYSEMA
Upon ascent, air escapes from a lung over pressurization into the tissues beneath the skin of the neck. It can be associated with mediastinal emphysema or can occur alone
CAUSES
– Holding breath during ascent while breathing compressed air – Lung disease causing air trapping – Diving with cold or chest congestion – Airway obstruction from foreign object in the mouth (gum, etc.)
SYMPTOMS
– Feeling of fullness in the neck area – Swelling or inflation around the neck and upper chest – Crackling sensation when skin is pressed – Change in sound of voice – Cough
TREATMENT
– Transport to nearest medical facility – ABC's - airway, breathing, and circulation – Administer 100% oxygen with diver in supine position – Monitor for shock – Unless showing signs of a gas embolism as well, recompression is not normally required

Diving Emergency Action Plan

LUNG OVER-EXPANSION INJURIES

MEDIASTINAL EMPHYSEMA
Upon ascent, air escapes from a lung over pressurization into the tissues surrounding the heart, major blood vessels, and trachea (windpipe). This gas expands on ascent, causing pain under the sternum (breastbone), shortness of breath, or in extreme cases, fainting from impaired blood return to the heart.
CAUSES
– Holding breath during ascent while breathing compressed air – Lung disease causing air trapping – Diving with cold or chest congestion – Airway obstruction from foreign object in the mouth (gum, etc.)
SYMPTOMS
– Difficulty breathing – Shortness of breath – Faintness – Pain under breastbone that may radiate to the neck, collarbone, or shoulder – Cyanosis (blueness) of the skin, lips, or nailbeds – Shock – Swelling around the neck – A brassy quality to the voice – A sensation of pressure on the windpipe – Cough – Deviation of Adam's apple to affected side
TREATMENT
– Transport to nearest medical facility – Mediastinal emphysema causing respiratory or circulatory impairment may require recompression – ABC's - airway, breathing, and circulation – Administer 100% oxygen – Monitor for shock

Diving Emergency Action Plan

DECOMPRESSION INJURIES

DECOMPRESSION SICKNESS (DCS)
The result of inadequate decompression following exposure to increased pressure. Typically occurs as a result of ascending too quickly and not allotting the proper amount of time required for decompression to occur. An important note is that DCS can occur even when divers follow dive tables and computers and remain within recommended decompression limits. Please listen to your body for the 24 hours following a dive and consult a physician familiar with dive medicine should you experience any of these symptoms
PREVENTION
– Make safety stops – Ascend slowly (maximum of 30 feet per minute) – Allow for longer surface intervals – Plan dives conservatively – Hydrate well when diving – Maintain good physical fitness and nutrition – Do not dive when dehydrated, intoxicated, hungover, or overly fatigued
SYMPTOMS
<i>* Symptoms often occur within 6 hours of dive completion, but can be delayed up to 24 hours</i> – Joint pain (most commonly in the elbow, shoulder, hip, or knee) – Extreme fatigue and weakness – Dizziness, tunnel vision, or staggering, occasionally leading to unconsciousness – Paralysis or numbness – Itchy skin and/or blotchy rash on the abdomen – Ringing in the ears or partial deafness, confusion, or disorientation
TREATMENT
– Transport to recompression chamber ASAP and consult with physician familiar with diving injuries * DO NOT attempt to treat in the water – Monitor ABC's - airway, breathing, and circulation – Administer 100% oxygen – Administer fluids (water) – Rest in supine position